Security Log Analysis Report

Detecting Failed SSH Authentication Attempts

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Tools used:

- Kali Linux
- Virtual Box
- OpenSSH
- journalctl
- grep
- wc

1. Introduction

This lab demonstrates the detection and analysis of failed SSH authentication attempts on a Linux system. The objective was to simulate suspicious login activity and analyze system logs to identify evidence of unauthorized access attempts.

The lab was conducted in a virtual machine environment using Kali Linux. Log analysis was performed using the systemd journal via the journalctl command.

2. Lab Environment

- Operating System: Kali Linux (Virtual Machine)
- Service Analyzed: OpenSSH (sshd)
- Log Source: systemd journal
- Tools Used:
    - systemctl
    - journalctl
    - grep
    - wc

3. Objective

The objective of this lab was to:

- Verify that the SSH service is active
- Simulate multiple failed login attempts
- Analyze authentication logs
- Filter and identify failed login entries
- Quantify suspicious login attempts

4. Methodology

Step 1: Verify SSH Service Status

The SSH service was verified using:

sudo systemctl status ssh

The service was confirmed to be active and running.

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
     Active: active (running) since Fri 2026-02-27 17:46:25 EST; 15min ago
 Invocation: e0fe21a2e7f14ab4815adc7f9e895bd7
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 15674 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 15676 (sshd)
      Tasks: 1 (limit: 2116)
     Memory: 4.6M (peak: 8.7M)
        CPU: 441ms
     CGroup: /system.slice/ssh.service
             └─15676 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 27 17:56:04 kali sshd-session[20304]: Invalid user fakeuser from 127.0.0.1 port 54788
Feb 27 17:56:07 kali sshd-session[20304]: pam_unix(sshd:auth): check pass; user unknown
Feb 27 17:56:07 kali sshd-session[20304]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Feb 27 17:56:09 kali sshd-session[20304]: Failed password for invalid user fakeuser from 127.0.0.1 port 54788 ssh2
Feb 27 17:56:11 kali sshd-session[20304]: pam_unix(sshd:auth): check pass; user unknown
Feb 27 17:56:13 kali sshd-session[20304]: Failed password for invalid user fakeuser from 127.0.0.1 port 54788 ssh2
Feb 27 17:56:16 kali sshd-session[20304]: pam_unix(sshd:auth): check pass; user unknown
Feb 27 17:56:19 kali sshd-session[20304]: Failed password for invalid user fakeuser from 127.0.0.1 port 54788 ssh2
Feb 27 17:56:21 kali sshd-session[20304]: Connection closed by invalid user fakeuser 127.0.0.1 port 54788 [preauth]
Feb 27 17:56:21 kali sshd-session[20304]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1

┌──(kali㉿kali)-[~]
└─$
```

Step 2: Simulate Failed Login Attempts

Multiple failed login attempts were generated using:

ssh fakeuser@127.0.0.1

A non-existent user (“fakeuser”) was used to simulate unauthorized access attempts. Random passwords were entered multiple times to generate authentication failures.

```
┌──(kali㉿kali)-[~]
└─$ ssh fakeuser@127.0.0.1
fakeuser@127.0.0.1's password:
Permission denied, please try again.
fakeuser@127.0.0.1's password:
Permission denied, please try again.
fakeuser@127.0.0.1's password:
fakeuser@127.0.0.1: Permission denied (publickey,password).

┌──(kali㉿kali)-[~]
└─$
```

Step 3: Analyze SSH Logs

System logs were reviewed using:

sudo journalctl -u ssh | tail -n 30

The output displayed authentication failures including:

- Failed password
- pam_unix(sshd:auth)
- authentication failure
- connection closed

```
┌──(kali㉿kali)-[~]
└─$ sudo journalctl -u ssh | tail -n 30
Feb 27 17:48:41 kali sshd-session[16548]: Failed password for invalid user fakeuser from ::1 port 58120 ssh2
Feb 27 17:49:04 kali sshd-session[16548]: pam_unix(sshd:auth): check pass; user unknown
Feb 27 17:49:06 kali sshd-session[16548]: Failed password for invalid user fakeuser from ::1 port 58120 ssh2
Feb 27 17:49:06 kali sshd-session[16548]: Connection closed by invalid user fakeuser ::1 port 58120 [preauth]
Feb 27 17:49:06 kali sshd-session[16548]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=::1
Feb 27 17:49:06 kali sshd-session[16953]: Invalid user fakeuser from ::1 port 36082
Feb 27 17:49:15 kali sshd-session[16953]: pam_unix(sshd:auth): check pass; user unknown
Feb 27 17:49:15 kali sshd-session[16953]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=::1
Feb 27 17:49:17 kali sshd-session[16953]: Failed password for invalid user fakeuser from ::1 port 36082 ssh2
Feb 27 17:49:54 kali sshd-session[16953]: Connection closed by invalid user fakeuser ::1 port 36082 [preauth]
Feb 27 17:56:04 kali sshd-session[20304]: Invalid user fakeuser from 127.0.0.1 port 54788
Feb 27 17:56:07 kali sshd-session[20304]: pam_unix(sshd:auth): check pass; user unknown
Feb 27 17:56:07 kali sshd-session[20304]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Feb 27 17:56:09 kali sshd-session[20304]: Failed password for invalid user fakeuser from 127.0.0.1 port 54788 ssh2
Feb 27 17:56:11 kali sshd-session[20304]: pam_unix(sshd:auth): check pass; user unknown
Feb 27 17:56:13 kali sshd-session[20304]: Failed password for invalid user fakeuser from 127.0.0.1 port 54788 ssh2
Feb 27 17:56:16 kali sshd-session[20304]: pam_unix(sshd:auth): check pass; user unknown
Feb 27 17:56:19 kali sshd-session[20304]: Failed password for invalid user fakeuser from 127.0.0.1 port 54788 ssh2
Feb 27 17:56:21 kali sshd-session[20304]: Connection closed by invalid user fakeuser 127.0.0.1 port 54788 [preauth]
Feb 27 17:56:21 kali sshd-session[20304]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Feb 27 18:04:32 kali sshd-session[24432]: Invalid user fakeuser from 127.0.0.1 port 33752
Feb 27 18:04:35 kali sshd-session[24432]: pam_unix(sshd:auth): check pass; user unknown
Feb 27 18:04:35 kali sshd-session[24432]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1
Feb 27 18:04:37 kali sshd-session[24432]: Failed password for invalid user fakeuser from 127.0.0.1 port 33752 ssh2
Feb 27 18:04:39 kali sshd-session[24432]: pam_unix(sshd:auth): check pass; user unknown
Feb 27 18:04:40 kali sshd-session[24432]: Failed password for invalid user fakeuser from 127.0.0.1 port 33752 ssh2
Feb 27 18:04:43 kali sshd-session[24432]: pam_unix(sshd:auth): check pass; user unknown
Feb 27 18:04:45 kali sshd-session[24432]: Failed password for invalid user fakeuser from 127.0.0.1 port 33752 ssh2
Feb 27 18:04:45 kali sshd-session[24432]: Connection closed by invalid user fakeuser 127.0.0.1 port 33752 [preauth]
Feb 27 18:04:45 kali sshd-session[24432]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=127.0.0.1

┌──(kali㉿kali)-[~]
└─$
```

Step 4: Filter Failed Authentication Events

To isolate only failed login attempts, the following command was used:

sudo journalctl -u ssh | grep "Failed"

This filtered the logs to display only entries related to failed password attempts.

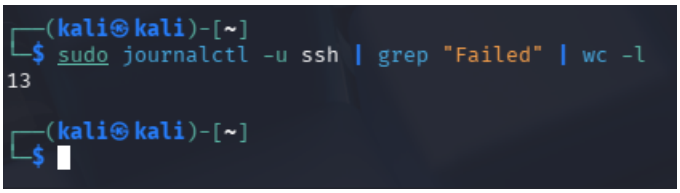
```
┌──(kali㉿kali)-[~]
└─$ sudo journalctl -u ssh | grep "Failed"
Feb 27 17:48:02 kali sshd-session[16145]: Failed password for invalid user fakeuser from ::1 port 48744 ssh2
Feb 27 17:48:09 kali sshd-session[16145]: Failed password for invalid user fakeuser from ::1 port 48744 ssh2
Feb 27 17:48:15 kali sshd-session[16145]: Failed password for invalid user fakeuser from ::1 port 48744 ssh2
Feb 27 17:48:24 kali sshd-session[16548]: Failed password for invalid user fakeuser from ::1 port 58120 ssh2
Feb 27 17:48:41 kali sshd-session[16548]: Failed password for invalid user fakeuser from ::1 port 58120 ssh2
Feb 27 17:49:06 kali sshd-session[16548]: Failed password for invalid user fakeuser from ::1 port 58120 ssh2
Feb 27 17:49:17 kali sshd-session[16953]: Failed password for invalid user fakeuser from ::1 port 36082 ssh2
Feb 27 17:56:09 kali sshd-session[20304]: Failed password for invalid user fakeuser from 127.0.0.1 port 54788 ssh2
Feb 27 17:56:13 kali sshd-session[20304]: Failed password for invalid user fakeuser from 127.0.0.1 port 54788 ssh2
Feb 27 17:56:19 kali sshd-session[20304]: Failed password for invalid user fakeuser from 127.0.0.1 port 54788 ssh2
Feb 27 18:04:37 kali sshd-session[24432]: Failed password for invalid user fakeuser from 127.0.0.1 port 33752 ssh2
Feb 27 18:04:40 kali sshd-session[24432]: Failed password for invalid user fakeuser from 127.0.0.1 port 33752 ssh2
Feb 27 18:04:45 kali sshd-session[24432]: Failed password for invalid user fakeuser from 127.0.0.1 port 33752 ssh2

┌──(kali㉿kali)-[~]
└─$
```

Step 5: Count Failed Login Attempts

To quantify the suspicious activity, failed attempts were counted using:

sudo journalctl -u ssh | grep "Failed" | wc -l

A total of 13 failed login attempts were detected.



5. Findings

Analysis of the SSH logs revealed multiple failed authentication attempts targeting a non-existent user account. The logs included repeated “Failed password” entries generated by the sshd service.

The repeated authentication failures simulate a brute-force attack pattern, where an attacker attempts multiple passwords to gain unauthorized access.

Key Observations

- A total of 13 failed SSH login attempts were detected.
- The attempts targeted a non-existent user account.
- Authentication failures were properly logged by the system.
- No successful login occurred.

The detection of repeated authentication failures demonstrates the importance of continuous log monitoring. In a production environment, this pattern would warrant alert generation and potential investigation for brute-force attack activity.

6. Risk Analysis

Repeated failed login attempts may indicate:

- Brute-force attack attempts
- Credential stuffing attempts
- Automated scanning activity

If successful, such attacks could result in:

- Unauthorized system access
- Privilege escalation
- Data exfiltration
- System compromise

Monitoring authentication logs is critical for early detection of malicious activity.

7. Recommendations

To mitigate SSH brute-force risks, the following security measures are recommended:

1. Enable Fail2Ban to block repeated failed attempts
2. Disable password-based authentication
3. Implement SSH key-based authentication
4. Limit login attempts
5. Restrict SSH access by IP address
6. Regularly monitor authentication logs

These controls significantly reduce the risk of unauthorized access.